

3. Basic functions of the synapses and neurotransmitters (II)

Characteristics of Some of the More Important Small-Molecule Transmitters:

1. *Acetylcholine* is secreted by neurons in many areas of the nervous system but specifically by (1) the terminals of the large pyramidal cells from the motor cortex, (2) several different types of neurons in the basal ganglia, (3) the motor neurons that innervate the skeletal muscles, (4) the preganglionic neurons of the autonomic nervous system, (5) the postganglionic neurons of the parasympathetic nervous system, and (6) some of the postganglionic neurons of the sympathetic nervous system. In most instances, acetylcholine has an excitatory effect; however, it is known to have inhibitory effects at some peripheral parasympathetic nerve endings, such as inhibition of the heart by the vagus nerves.
2. *Norepinephrine* is secreted by the terminals of many neurons whose cell bodies are located in the brain stem and hypothalamus. Specifically, norepinephrine-secreting neurons located in the *locus ceruleus* in the pons send nerve fibers to widespread areas of the brain to help control overall activity and mood of the mind, such as increasing the level of wakefulness. In most of these areas, norepinephrine probably activates excitatory receptors, but in a few areas, it activates inhibitory receptors instead. Norepinephrine is also secreted by most postganglionic neurons of the sympathetic nervous system, where it excites some organs but inhibits others.
3. *Dopamine* is secreted by neurons that originate in the substantia nigra. The termination of these neurons is mainly in the striatal region of the basal ganglia. The effect of dopamine is usually inhibition.
4. *Glycine* is secreted mainly at synapses in the spinal cord. It is believed to always act as an inhibitory transmitter.

5. *GABA (gamma-aminobutyric acid)* is secreted by nerve terminals in the spinal cord, cerebellum, basal ganglia, and many areas of the cortex. It is believed always to cause inhibition.
6. *Glutamate* is secreted by the presynaptic terminals in many of the sensory pathways entering the central nervous system, as well as in many areas of the cerebral cortex. It probably always causes excitation.
7. *Serotonin* is secreted by nuclei that originate in the median raphe of the brain stem and project to many brain and spinal cord areas, especially to the dorsal horns of the spinal cord and to the hypothalamus. Serotonin acts as an inhibitor of pain pathways in the cord, and an inhibitor action in the higher regions of the nervous system is believed to help control the mood of the person, perhaps even to cause sleep.
8. *Nitric oxide* is especially secreted by nerve terminals in areas of the brain responsible for long-term behavior and for memory. Therefore, this transmitter system might in the future explain some behavior and memory functions that thus far have defied understanding. Nitric oxide is different from other small-molecule transmitters in its mechanism of formation in the presynaptic terminal and in its actions on the postsynaptic neuron. It is not preformed and stored in vesicles in the presynaptic terminal as are other transmitters. Instead, it is synthesized almost instantly as needed, and it then diffuses out of the presynaptic terminals over a period of seconds rather than being released in vesicular packets. Next, it diffuses into postsynaptic neurons nearby. In the postsynaptic neuron, it usually does not greatly alter the membrane potential but instead changes intracellular metabolic functions that modify neuronal excitability for seconds, minutes, or perhaps even longer.

Neuropeptides:

Neuropeptides are synthesized differently and have actions that are usually slow and in other ways quite different from those of the small-molecule transmitters. The neuropeptides are not synthesized in the cytosol of the presynaptic terminals. Instead, they are synthesized as integral parts of large-protein molecules by ribosomes in the neuronal cell body.

The protein molecules then enter the spaces inside the endoplasmic reticulum of the cell body and subsequently inside the Golgi apparatus, where two changes occur: First, the neuropeptide-forming protein is enzymatically split into smaller fragments, some of which are either the neuropeptide itself or a precursor of it. Second, the Golgi apparatus packages the neuropeptide into minute transmitter vesicles that are released into the cytoplasm. Then the transmitter vesicles are transported all the way to the tips of the nerve fibers by *axonal streaming* of the axon cytoplasm, traveling at the slow rate of only a few centimeters per day. Finally, these vesicles release their transmitter at the neuronal terminals in response to action potentials in the same manner as for small-molecule transmitters. However, the vesicle is autolyzed and is not reused.

Because of this laborious method of forming the neuropeptides, much smaller quantities of them are usually released than of the small-molecule transmitters. This is partly compensated for by the fact that the neuropeptides are generally a thousand or more times as potent as the small-molecule transmitters. Another important characteristic of the neuropeptides is that they often cause much more prolonged actions. Some of these actions include prolonged closure of calcium channels, prolonged changes in the metabolic machinery of cells, prolonged changes in activation or deactivation of specific genes in the cell nucleus, and/or prolonged alterations in numbers of excitatory or inhibitory receptors. Some of these effects last for days, but others perhaps for months or years. Our knowledge of the functions of the neuropeptides is only beginning to develop.

THE NEURONAL POOL (figure 10):

The arrangement of the functionally related neurons in the CNS takes the form of neuronal pool (NP). The NP show the following arrangements:

1. Divergence:

A single input neuron (presynaptic) synapses through its terminal branches with multiple output (postsynaptic) neurons.

Significance: It allows spreading of impulses from one single input neuron to multiple output neurons.

Example: stimulation of one single cortical neuron in the motor area of the CC can activate thousands of AHCs in the spinal cord resulting in contraction of multiple motor units of skeletal muscles.

2. Convergence:

- The terminal branches of multiple input neurons converge on one single output neuron.
- Significance: it allows for spatial summation whether excitatory or inhibitory to occur in the output neuron.
- Example: one single AHC receives multiple converging input fibers from different sites as follows:
 1. Afferent sensory fibers from the dorsal roots of the same and opposite side at the same level → mediate reflex responses (e.g. muscle tone, withdrawal reflex,...etc)
 2. Afferent sensory fibers from the dorsal roots of the same and opposite side at higher or lower levels through the propriospinal tract.
 3. Fibers from descending motor tracts (pyramidal and extrapyramidal) controlling motor functions.

3. After Discharge (Interneurone Barrage):

It means prolonged discharge from the output neuron after stoppage of discharge from the input neuron.

Mechanism:

A. Parallel Chain Circuit:

The input neuron stimulates the output neuron both directly and through several parallel chain circuits of IN that finally converge on the same output neuron. The greater the number of synapses in the parallel circuits, the more is the delay of arrival of impulses to the output neuron.

The impulses therefore, reach the output neuron in succession after stoppage of discharge from the input neuron → prolonged discharge from the output neuron.

B. Closed Chain (reverberatory) Circuit:

The input neuron stimulate the output neuron directly.

Collaterals from the output neuron stimulate multiple IN forming closed chain circuits that finally converge on the same output neuron restimulating it even after stoppage of input impulses.

Reverberation stops by:

Synaptic fatigue.

Synaptic inhibition from other sites.

Significance:

- ❑ It is an example of +ve feedback.
- ❑ In the facilitatory RF of the brain, it ensures the persistent discharge to keep the CC wakeful and alert for 14-18 h.
- ❑ It continuously stimulates the Ap C and IC in the brain stem. These are stopped by HB reflexes to stop inspiration and begin expiration.

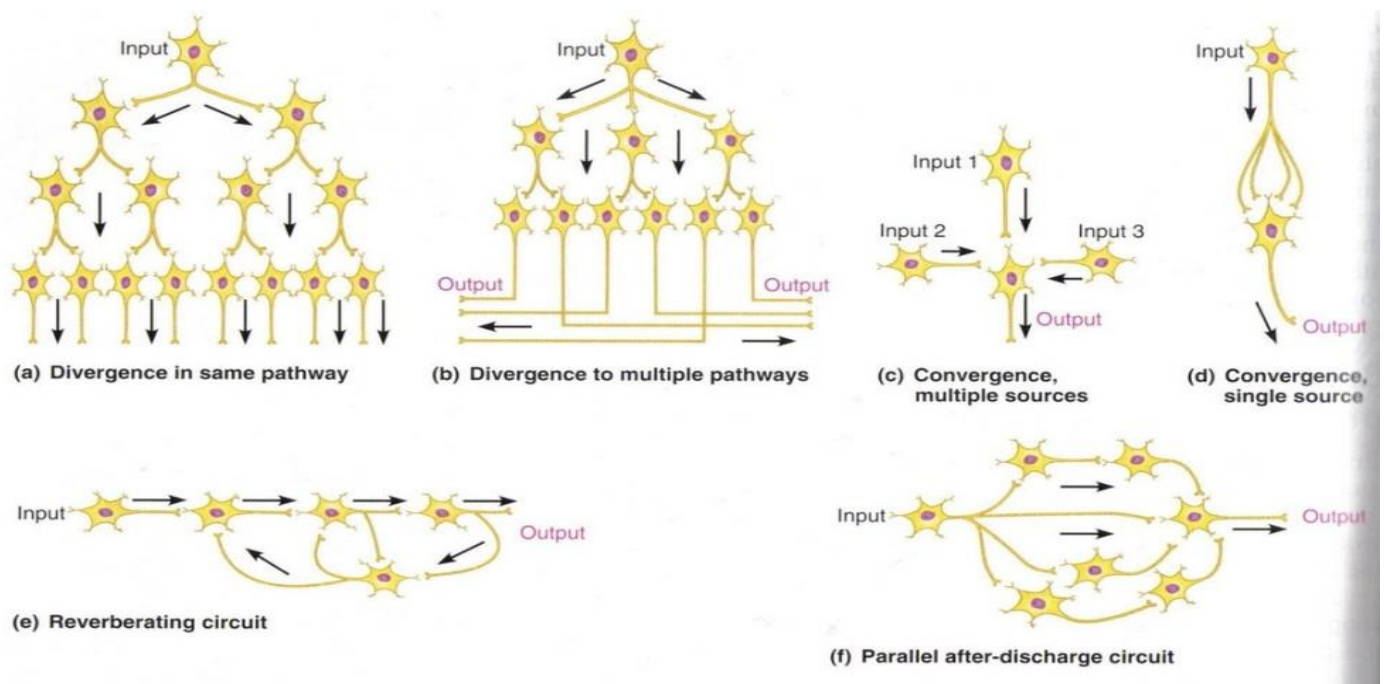


Figure 10: Neuronal Pool

4. Excitation Field (EF), Discharge Zone (DZ) and Subliminal Fringe (SF) (figure 11):
 - Excitation Field (EF): the nerve terminals of the input neuron end by synaptic knobs on a large number of efferent output neurons (the EF of the input neuron).
 - Discharge Zone (DZ): the neurons in the central area of the EF receive large number of terminal knobs from input neuron, therefore, when this input neuron is stimulated, spatial summation will allow prolonged APs in the output neurons (DZ).
 - Subliminal Fringe (SF): the neurons in the peripheral area of the EF receive small number of terminal knobs from input neuron, therefore, when this input neuron is stimulated, spatial summation will produce partial depolarization with increased excitability not sufficient to produce prolonged APs in the output neurons (SF).

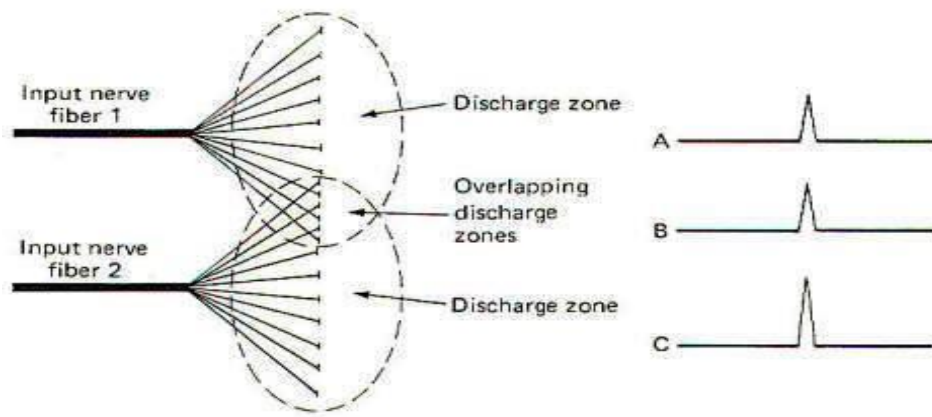


Figure 11: Discharge zone and subliminal fringe.

PROPERTIES OF SYNAPTIC TRANSMISSION:

1. Forward Conduction: Conduction occurs from presynaptic to postsynaptic neuron and never the reverse.
2. Synaptic Delay:
 - It is the time required for conduction of a nerve impulse from the presynaptic to the postsynaptic neuron.
 - For a single synapse, it is about 0.5 ms.
 - It is required for steps of synaptic transmission.
 - It is important to determine whether the reflex is monosynaptic or polysynaptic.
3. Synaptic Fatigue:
 - If the presynaptic knob is repeatedly stimulated at a high rate, the frequency of discharge in the postsynaptic neuron increases at first → then decreases rapidly → then stops in spite of persistent stimulation. This is called synaptic fatigue.
 - Causes: Depletion of presynaptic chemical transmitter due to high frequency stimulation (the rate of release > the rate of synthesis).

- Significance: it prevents over activity of the CNS.
- Example: tetanic convulsions of epileptic fits are followed by synaptic fatigue at the MEP → generalized muscle weakness.

4. Synaptic Potentiation:

- If the presynaptic knob is repeatedly stimulated for a short period (not sufficient to produce fatigue) → higher and prolonged EPSP → increased excitability of the postsynaptic neuron which remains for sometimes after stoppage of presynaptic stimulation. This is called synaptic (or post tetanic) potentiation.
- Cause: increased permeability to Ca^{2+} → ↑ presynaptic accumulation of Ca^{2+} → ↑ release of chemical transmitter → EPSP of two types depending on the ionic channels affected : 1) a fast short EPSP, 2) a slow prolonged EPSP → ↑ excitability for weaker stimuli that may last for seconds to hours or days.

5. Effect of Hypoxia:

- Synaptic transmission is absolutely dependent on the presence of O_2 .
- Hypoxia for few seconds will stop synaptic transmission e.g. interruption of cerebral circulation for 3-5 seconds will lead to unconsciousness due to failure of synaptic transmission in the CNS.

6. Effect of Acidosis and Alkalosis:

- Alkalosis: → precipitates Ca^{2+} → increased Na^+ influx into neurons → increased excitability and facilitates synaptic transmission → tetanic convulsions. Significance: hyperventilation → wash of CO_2 → alkalosis → epileptic fits in predisposed individuals.
- Acidosis: → produces the opposite effect → ↓ synaptic transmission → coma. Significance: uncontrolled DM → acidosis → ↓ synaptic transmission → diabetic coma.

7. Effect of Drugs (table 1):

Table 1: effect of drugs on synaptic transmission

Drugs that facilitate ST	Drugs that inhibit ST
Increasing the release of transmitter	Decreasing the release of transmitter
Changing the permeability of postsynaptic neuron to ions $\rightarrow \uparrow \text{Na}^+$ influx and $\downarrow \text{K}^+$ out flux $\rightarrow \uparrow \text{EPSP}$ and facilitate ST e.g. Caffeine.	Changing the permeability of postsynaptic neuron to ions $\rightarrow \downarrow \text{Na}^+$ influx and $\uparrow \text{K}^+$ out flux $\rightarrow \uparrow \text{IPSP}$ and inhibition of ST e.g. Anaesthetics.
Decreasing the release of inhibitory transmitters or blocking their effects e.g Strychnine .	Increasing the release of inhibitory transmitters.

CHEMICAL TRANSMITTERS:

Almost all the synapses used for signal transmission in the central nervous system of the human being are *chemical synapses*. In these, the first neuron secretes at its nerve ending synapse a chemical substance called a *neurotransmitter* (or often called simply *transmitter substance*), and this transmitter in turn acts on receptor proteins in the membrane of the next neuron to excite the neuron, inhibit it, or modify its sensitivity in some other way. More than 40 important transmitter substances have been discovered thus far. Some of the best known are acetylcholine, norepinephrine, epinephrine, histamine, gamma-aminobutyric acid (GABA), glycine, serotonin, and glutamate.

Chemical Substances That Function as Synaptic Transmitters (table 2):

More than 50 chemical substances have been proved or postulated to function as synaptic transmitters. Many of them are listed in (SEE Tables), which give two groups of synaptic transmitters. One group comprises *small-molecule, rapidly acting*

transmitters. The other is made up of a large number of *neuropeptides* of much larger molecular size that are usually much more slowly acting.

Table 2: Chemical Substances That Function as Synaptic Transmitters

Class I Acetylcholine	Hypothalamic-releasing hormones	Peptides that act on gut and brain
Class II: The Amines Norepinephrine Epinephrine Dopamine Serotonin Histamine	Thyrotropin-releasing hormone Luteinizing hormone–releasing hormone Somatostatin (growth hormone inhibitory factor)	Leucine enkephalin Methionine enkephalin Substance P Gastrin Cholecystokinin
Class III: Amino Acids Gamma-aminobutyric acid (GABA) Glycine Glutamate Aspartate	Pituitary peptides Adrenocorticotrophic hormone (ACTH)	Vasoactive intestinal polypeptide (VIP) Nerve growth factor Brain-derived neurotropic factor
Class IV Nitric oxide (NO)	β -Endorphin α -Melanocyte-stimulating hormone Prolactin Luteinizing hormone Thyrotropin Growth hormone Vasopressin	Insulin Glucagon From other tissues Angiotensin II Bradykinin Carnosine

	Oxytocin	Sleep peptides Calcitonin
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- The small-molecule, rapidly acting transmitters are the ones that cause most acute responses of the nervous system, such as transmission of sensory signals to the brain and of motor signals back to the muscles.
- The neuropeptides, in contrast, usually cause more prolonged actions, such as long-term changes in numbers of neuronal receptors, long-term opening or closure of certain ion channels, and possibly even long-term changes in numbers of synapses or sizes of synapses.

Small-Molecule, Rapidly Acting Transmitters:

In most cases, the small-molecule types of transmitters are synthesized in the cytosol of the presynaptic terminal and are absorbed by means of active transport into the many transmitter vesicles in the terminal. Then, each time an action potential reaches the presynaptic terminal, a few vesicles at a time release their transmitter into the synaptic cleft.

This usually occurs within a millisecond or less by the mechanism described earlier. The subsequent action of the small-molecule type of transmitter on the membrane receptors of the postsynaptic neuron usually also occurs within another millisecond or less. Most often the effect is to increase or decrease conductance through ion channels; an example is to increase sodium conductance, which causes excitation, or to increase potassium or chloride conductance, which causes inhibition.

Recycling of the Small-Molecule Types of Vesicles:

Vesicles that store and release small-molecule transmitters are continually recycled and used over and over again. After they fuse with the synaptic membrane and open to release their transmitter substance, the vesicle membrane at first simply becomes part of the synaptic membrane. However, within seconds to minutes, the

vesicle portion of the membrane invaginates back to the inside of the presynaptic terminal and pinches off to form a new vesicle. And the new vesicular membrane still contains appropriate enzyme proteins or transport proteins required for synthesizing and/or concentrating new transmitter substance inside the vesicle.

Acetylcholine is a typical small-molecule transmitter that obeys the principles of synthesis and release stated earlier. This transmitter substance is synthesized in the presynaptic terminal from acetyl coenzyme A and choline in the presence of the enzyme *choline acetyltransferase*. Then it is transported into its specific vesicles. When the vesicles later release the acetylcholine into the synaptic cleft during synaptic neuronal signal transmission, the acetylcholine is rapidly split again to acetate and choline by the enzyme *cholinesterase*, which is present in the proteoglycan reticulum that fills the space of the synaptic cleft. And then again, inside the presynaptic terminal, the vesicles are recycled; choline is actively transported back into the terminal to be used again for synthesis of new acetylcholine.